

BT513T

Assignment 2 Fall

24

Opening Date: 11/11/2024

Closing Date: 15/11/2024

Total Marks: 10

Question No. 1

EXPLAIN THE GENERAL REQUIREMENTS OF FERMENTERS.

Fermenters are specialized bioreactors used for the growth of microorganisms, cells, or enzymes under controlled conditions to produce desired products, such as antibiotics, enzymes, and biofuels. The general requirements of fermenters include:

1. Sterility

- Preventing contamination is crucial. Fermenters should be designed to allow sterilization, usually by autoclaving or steam-in-place (SIP) methods, to maintain sterile conditions throughout the fermentation process.

2. Control of Environmental Conditions

- **Temperature:** Fermenters should have temperature control systems (heating/cooling jackets) to maintain the optimum temperature for microbial or cell growth.
- **pH:** pH probes and control mechanisms (such as acid or base addition) are needed to maintain the desired pH, as different organisms require specific pH ranges.
- **Oxygen Supply:** Aerobic processes need efficient oxygen transfer. Thus, fermenters are equipped with spargers or diffusers to introduce air or oxygen, and agitators to enhance oxygen dissolution in the medium.

3. Agitation and Mixing

- Proper mixing ensures uniform distribution of nutrients, heat, and oxygen, preventing gradients within the fermenter. Agitators or impellers provide this mixing, and the speed can be controlled based on process requirements.

4. Nutrient Supply

- Fermenters need a continuous or controlled supply of nutrients (such as carbon, nitrogen, and minerals) for optimal growth and productivity. This often includes ports for feeding additional nutrients during the process.

5. Waste Gas Removal

- Gases like CO₂, produced during fermentation, should be removed to prevent accumulation that can inhibit growth or productivity. Exhaust gas systems with filters help in removing and managing waste gases.

6. Monitoring and Control Systems

- Sensors for temperature, pH, dissolved oxygen, and other parameters are essential for real-time monitoring. Automated control systems help maintain optimal conditions by adjusting variables like aeration, agitation, and nutrient feed.

7. Scale-Up Capability

- Industrial fermenters should be scalable to facilitate larger production volumes, maintaining similar control and conditions across various scales from lab to production.

8. Material Compatibility

- Fermenters are usually made of stainless steel or glass to resist corrosion, allow easy cleaning, and withstand sterilization. The material should be non-toxic and compatible with the fermentation media and cleaning agents.

9. Harvesting and Sampling Ports

- Ports for sampling allow periodic checks on the fermentation process. Harvesting systems enable easy extraction of the final product or cells without contaminating the system.

10. Compliance with Safety Standards

- Fermenters must meet safety standards, particularly for handling genetically modified organisms (GMOs) or toxic by-products, to protect workers and the environment.

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